

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Domain and Range

Example (Domain from an equation):

$f(x) = \sqrt{x-2}$ needs $x-2 \geq 0$, so the domain is $x \geq 2$.

Directions: State the domain. Show the inequality you solved.

1. $f(x) = \frac{1}{x-5}$

4. $f(x) = \frac{x}{x^2-9}$

2. $f(x) = \sqrt{x+3}$

5. $f(x) = \sqrt{10-x}$

3. $f(x) = x^2 + 1$

Challenge Problems

6. $f(x) = \frac{\sqrt{x-1}}{x-4}$ — give the domain (two restrictions).

Answer Key and Procedure

Procedure

1. Set each denominator $\neq 0$; solve.
2. Set each radicand ≥ 0 ; solve.
3. Combine the results for the domain.

Answers

1. $x \neq 5$
2. $x \geq -3$
3. all real numbers
4. $x \neq 3, x \neq -3$
5. $x \leq 10$
6. $x \geq 1$ and $x \neq 4$